KU LEUVEN

FACULTY OF
ENGINEERING SCIENCE

Decoding selective auditory attention from EEG to speech using contrastive learning with transformers

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Master of Science in
Biomedical Engineering, option
Bio-Informatics and AI

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Preface

Chloé Dirix

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Abstract

This `abstract` environment provides a summary of the work, which may or may not be extensive. The intention is that this should be limited to 1 page.

List of Figures and Tables

List of Figures

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List of Tables

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List of abbreviations and symbols

abbreviations

LoG	Laplacian-of-Gaussian
MSE	Mean Square error
PSNR	Peak Signal-to-Noise ratio

Symbols

42	"The Answer to the Ultimate Question of Life, the Universe, and Everything" volgens de [1]
c	Speed of light
E	Energy
m	Mass
π	The number pi

Chapter 1

Introduction

The first contains a general introduction to the work. The goals are defined and the modus operandi is explained.

1.1 background

Chapter 2

The First Chapter

A chapter is a logical unit. It normally starts with an introduction, which you are reading now. The last topic of the chapter holds the conclusion.

2.1 The First Topic of the Chapter

First comes the introduction to this topic.

2.1.1 An item

Please don't abuse enumerations: short enumerations shouldn't use "`itemize`" or "`enumerate`" environments. So *never write*:

> The Eiffel tower has three floors:
>
> - the first one;
> - the second one;
> - the third one.

But write:

> The Eiffel tower has three floors: the first one, the second one, and the third one.

2.2 A Second Topic

First we start with equation 2.1 after this paragraph.

$$A = \pi r^2 \tag{2.1}$$

2.3 Conclusion

The final section of the chapter gives an overview of the important results of this chapter. This implies that the introductory chapter and the concluding chapter don't need a conclusion.

Chapter 3

The Next Chapter

3.1 The First Topic of this Chapter

3.1.1 An item

A master's thesis is never an isolated work. This means that your text must contain references. On-line documents[3] as well as books[2] can be referenced.

Usually a text also contains inline formulas ($\sin^2\eta + \cos^2\eta = 1$) or formulas as separate equations.

$$\sigma(t) = \frac{1}{\sqrt{2\pi}} \int_0^t e^{-x^2/2} dx \tag{3.1}$$

3.2 Figures

Figures are used to add illustrations to the text. The Figure 3.1 shows the KU Leuven logo as an illustration.

FIGURE 3.1: The logo of the Faculty of Engineering Science.

gnats	gram	$13.65
	each	.01
gnu	stuffed	92.50
emu		33.33
armadillo	frozen	8.99

Table 3.1: A table with the wrong layout.

Item		
Animal	Description	Price ($)
Gnat	per gram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

Table 3.2: A table with the correct layout.

3.3 Tables

Tables are used to present data neatly arranged. A table is normally not a spreadsheet! Compare Table 3.1 en Table 3.2: which table do you prefer?

3.4 Conclusion

The final section of the chapter gives an overview of the important results of this chapter. This implies that the introductory chapter and the concluding chapter don't need a conclusion.

Chapter 4

the Loss Function

Embeddings are represented as:

$$\begin{aligned} \mathbf{e}_i &\in \mathbb{R}^D \quad \text{(EEG embedding)} \\ \mathbf{sa}_i &\in \mathbb{R}^D \quad \text{(attended speech embedding)} \\ \mathbf{su}_i &\in \mathbb{R}^D \quad \text{(unattended speech embedding)} \end{aligned}$$

Step 1: compute the similarity matrix

To compare the embeddings in the latent space the measure of similarity is used. The similarity between two embeddings is defined as their inner product divided by a learnable temperature parameter τ.

$$\text{sim}(\mathbf{e_i}, \mathbf{s_i}) = \frac{< \mathbf{e_i}, \mathbf{s_i} >}{\tau}$$

When computing all possible combinations, two matrices are created. S_A contains the matches between the eeg embeddings and the attended speech embeddings. S_U contains eeg-unattended speech combinations.

$$S_A = \begin{bmatrix} \text{sim}\big(\mathbf{e}_0, \mathbf{sa}_0\big) & \cdots & \text{sim}\big(\mathbf{e}_0, \mathbf{sa}_{N-1}\big) \\ \vdots & \ddots & \vdots \\ \text{sim}\big(\mathbf{e}_{N-1}, \mathbf{sa}_0\big) & \cdots & \text{sim}\big(\mathbf{e}_{N-1}, \mathbf{sa}_{N-1}\big) \end{bmatrix}$$

$$S_U = \begin{bmatrix} \text{sim}\big(\mathbf{e}_0, \mathbf{su}_0\big) & \cdots & \text{sim}\big(\mathbf{e}_0, \mathbf{su}_{N-1}\big) \\ \vdots & \ddots & \vdots \\ \text{sim}\big(\mathbf{e}_{N-1}, \mathbf{su}_0\big) & \cdots & \text{sim}\big(\mathbf{e}_{N-1}, \mathbf{su}_{N-1}\big) \end{bmatrix}$$

The diagonal of S_A corresponds to correctly aligned EEG–attended speech pairs. The diagonal of S_U corresponds to aligned unattended speech, which is treated separately as a weak negative. Only S_A is used in the CLIP loss computation. S_U is excluded from the contrastive classification and is used solely in the weak-negative term. This means S_U will in practice only need to be diagonal, as the rest of the combinations will not be used.

Step 2: compute the softmax probabilities

For a given EEG embedding $\mathbf{e}_i$, the softmax probability p_{e_i} measures how likely the attended speech $\mathbf{sa}_i$ is relative to all other attended speech candidates in the batch. This formulation treats mismatched attended speech as contrastive negatives.

$$p_{e_i} = \frac{\exp(S_A(i,i))}{\sum_{j=1}^{B} \exp(S_A(i,j))}$$

For symmetry, we also compute the probability that $\mathbf{e}_i$ is the correct EEG match for $\mathbf{sa}_i$:

$$p_{sa_i} = \frac{\exp\big(\mathrm{sim}(\mathbf{sa}_i, \mathbf{e}_i)/\tau\big)}{\sum_{j=1}^{B} \exp\big(\mathrm{sim}(\mathbf{sa}_i, \mathbf{e}_j)/\tau\big)}$$

Step 3: Compute the the symmetric CLIP loss

The loss is defined as:

$$\mathcal{L}_e = -\frac{1}{B}\sum_{i=1}^{B} \log p_{e_i}$$

$$\mathcal{L}_{sa} = -\frac{1}{B}\sum_{i=1}^{B} \log p_{sa_i}$$

$$\mathcal{L}_{CLIP} = \frac{1}{2}\Big(\mathcal{L}_e + \mathcal{L}_{sa}\Big)$$

Step 4: Introducing the weak negative

The temporally aligned unattended speech is treated as a weak negative using a margin ranking term. This margin-based term enforces that the EEG embedding is more similar to the attended speech than to the temporally aligned unattended speech by at least a margin m. Unlike the CLIP loss, this term does not force unattended speech to be far from the EEG embedding, but only enforces a relative ordering, reflecting partial neural tracking of unattended speech.

$$\mathcal{L}_{\text{weak}} = \frac{1}{B}\sum_{i=1}^{B} \max\Big(0,\ m - \big(\mathrm{sim}(\mathbf{e}_i, \mathbf{sa}_i) - \mathrm{sim}(\mathbf{e}_i, \mathbf{su}_i)\big)\Big).$$

Finally, the total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{CLIP} + \lambda \mathcal{L}_{\text{weak}}$$

In summary, the symmetric CLIP loss learns a shared embedding space by aligning EEG with attended speech using in-batch negatives, while the weak-negative term introduces task-specific structure by enforcing that attended speech is more similar to EEG than the temporally aligned unattended speech. This separation avoids treating unattended speech as a hard negative, which would contradict known neurophysiological evidence that EEG partially tracks both speech streams.

Chapter 5

Conclusion

The final chapter contains the overall conclusion. It also contains suggestions for future work and industrial applications.

Appendices

Appendix A

The First Appendix

Appendices hold useful data which is not essential to understand the work done in the master's thesis. An example is a (program) source. An appendix can also have sections as well as figures and references[1].

A.1 First appendix

Appendix B

The Last Appendix

Appendices are numbered with letters, but the sections and subsections use arabic numerals, as can be seen below.

Bibliography

[1] D. Adams. *The Hitchhiker's Guide to the Galaxy.* Del Rey (reprint), 1995. ISBN-13: 978-0345391803.

[2] T. Pratchett and N. Gaiman. *Good Omens:* The Nice and Accurate Prophecies of Agnes Nutter, Witch. HarperTorch (reprint), 2006. ISBN-13: 978-0060853983.

[3] Wikipedia. Thesis or dissertation. URL: http://en.wikipedia.org/wiki/Thesis_or_dissertation, last checked on 2010-01-07.

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